Assignment Week 3 – Differential Gene Expression Analysis

Case Study: GSE10072 – Lung Adenocarcinoma

1. Introduction

   Lung cancer remains one of the leading causes of cancer-related mortality worldwide. Among its subtypes, lung adenocarcinoma is the most common form, originating from epithelial cells in the distal airways. Tumorigenesis is driven by widespread molecular alterations, including dysregulation of gene expression affecting cell proliferation, apoptosis, and cell cycle control.

   Transcriptomic profiling using microarray technology enables comprehensive analysis of gene expression changes between cancerous and normal tissues. The dataset GSE10072, available from the Gene Expression Omnibus, provides gene expression data comparing lung adenocarcinoma samples and normal lung tissues.

   This study aims to:

   1) Identify significantly upregulated and downregulated genes.
   2) Visualize Differentially Expressed Genes (DEGs) using a volcano plot.
   3) Display the top 50 DEGs in a heatmap.
   4) Perform functional enrichment analysis using:
      - Gene Ontology (GO)
      - KEGG Pathway

2. Methods

   2.1 Differential Gene Expression

   The volcano plot revealed a substantial number of genes significantly upregulated and downregulated in lung adenocarcinoma samples compared to normal lung tissues.

   Upregulated genes were primarily associated with cell proliferation and cell cycle progression, while downregulated genes were often related to normal lung physiological functions.

   2.2 Top 50 Differentially Expressed Genes

   The heatmap of the top 50 DEGs demonstrated clear clustering between cancer and normal samples. Cancer samples formed a distinct cluster, indicating a strong gene expression signature characteristic of lung adenocarcinoma.

   2.3 Gene Ontology (GO) Enrichment

   GO enrichment analysis identified significant biological processes including: 1) Cell cycle regulation; 2) DNA replication; 3) Mitotic nuclear division; and 4) Chromosome segregation

   These processes reflect enhanced proliferative activity in tumor cells.

   2.4 KEGG Pathway Enrichment

   KEGG pathway analysis highlighted pathways such as: 1) Cell cycle; 2) p53 signaling pathway; 3) DNA replication; and 4) Pathways in cancer

   These pathways are consistent with known molecular mechanisms driving lung adenocarcinoma progression.

3. Results and Interpretation

3.1 Differential Gene Expression Analysis

Differential expression analysis was performed to compare lung adenocarcinoma tissues with normal lung samples. A total of 646 genes were identified as significantly differentially expressed (adjusted p-value < 0.05 and |log2FC| > 1), consisting of:

297 upregulated genes; and

349 downregulated genes

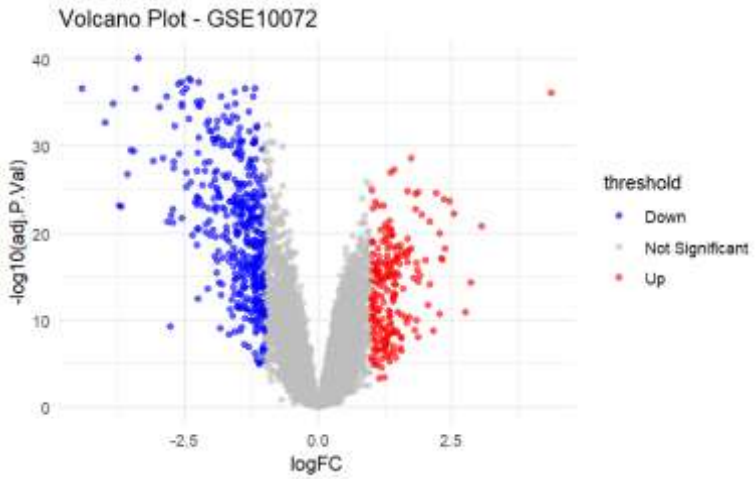


Fig 1. Volcano Plot

The volcano plot demonstrated a clear separation between significantly upregulated and downregulated genes, indicating substantial transcriptomic alterations in tumor tissues. The distribution pattern suggests widespread molecular dysregulation in lung adenocarcinoma. Upregulated genes were predominantly associated with: 1) Cell cycle progression; 2) Mitotic division; 3) DNA replication; and 4) Chromosomal segregation. These findings are consistent with the hallmark feature of cancer: uncontrolled cellular proliferation. Conversely, downregulated genes were mainly related to: 1) Normal lung physiological processes; 2) Tissue structural maintenance; and 3) Differentiation-associated pathways. This indicates loss of normal cellular identity and functional disruption in tumor tissues.

3.2 Clustering of Top 50 Differentially Expressed Genes

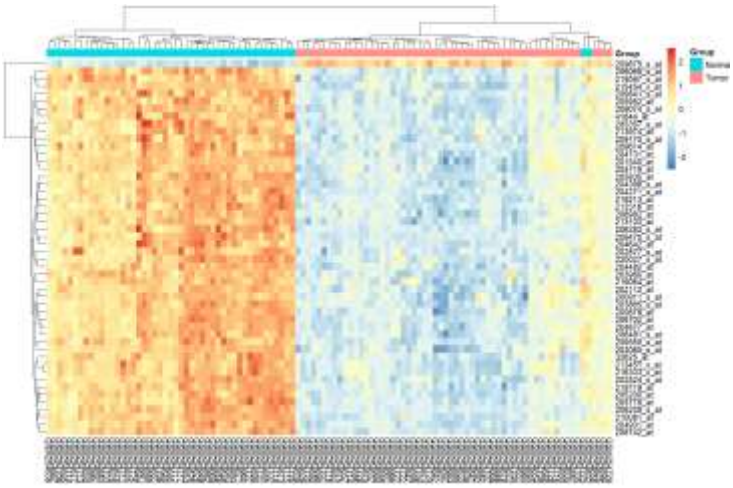
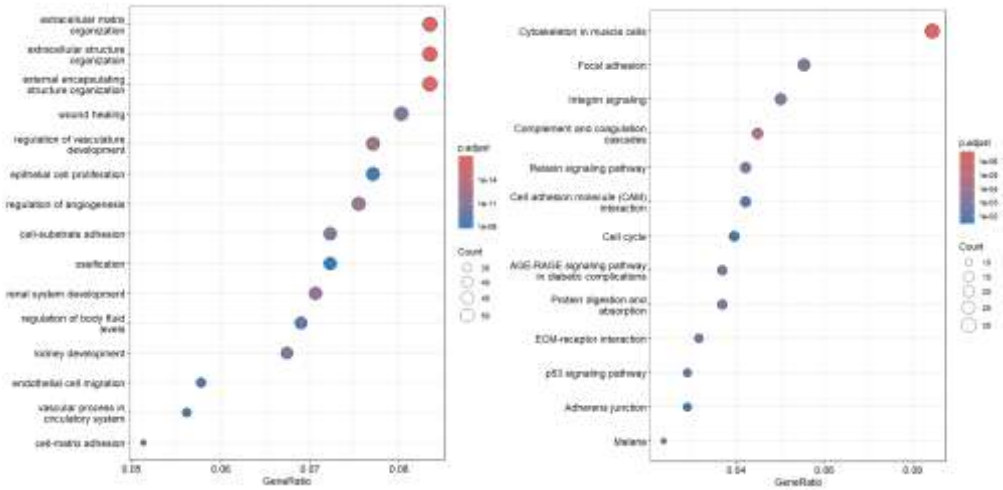


Fig 2. Heatmap

A heatmap was generated using the top 50 most significant DEGs. Hierarchical clustering revealed a clear segregation between tumor and normal samples and a distinct gene expression patterns characteristic of lung adenocarcinoma. Tumor samples clustered tightly together, reflecting a consistent oncogenic transcriptional signature. Normal samples formed a separate cluster, confirming strong molecular differences between the two biological conditions. This clustering pattern supports the robustness of the differential expression results.

3.3 Gene Ontology (GO) Analysis



GO Biological Process enrichment analysis revealed significant enrichment in cell cycle regulation, DNA replication, mitotic nuclear division, and chromosome segregation. These processes directly reflect enhanced proliferative activity in tumor cells. Dysregulation of the cell cycle machinery is a central driver of tumorigenesis, allowing continuous cell division independent of regulatory checkpoints. The enrichment of mitotic-related processes further supports the presence of uncontrolled cell division in lung adenocarcinoma tissues.

3.4 KEGG Pathway Enrichment Analysis

KEGG pathway analysis identified several significantly enriched pathways: 1) Cell cycle pathway; 2) p53 signalling pathway; 3) DNA replication; and 4) Pathways in cancer. The activation of the cell cycle pathway reinforces findings from GO analysis. The enrichment of the p53 signalling pathway is particularly important, as p53 plays a critical role in tumor suppression by regulating DNA repair, apoptosis, and cell cycle arrest. Disruption of this pathway is a well-established mechanism in lung adenocarcinoma development. The "Pathways in cancer" category integrates multiple oncogenic signalling cascades, confirming that the identified DEGs are functionally linked to known cancer-driving mechanisms.

4. Conclusion

This transcriptomic analysis of GSE10072 successfully identified significant gene expression differences between lung adenocarcinoma and normal lung tissues. Several key findings include:

1. Identification of 646 significant DEGs
2. Extensive upregulation of genes involved in cell proliferation and mitotic processes
3. Significant enrichment of cell cycle and DNA replication pathways
4. Activation of oncogenic signalling pathways including p53-related mechanisms
5. Clear molecular separation between tumor and normal samples based on hierarchical clustering

Overall, this study demonstrates that lung adenocarcinoma is characterized by profound dysregulation of genes governing cell cycle control and genomic stability. The integration of differential expression analysis with functional enrichment provides valuable insight into the molecular mechanisms underlying tumor development and progression. These findings underscore the power of transcriptomic profiling in identifying potential molecular targets and improving our understanding of lung cancer biology.